

Editor-in-Chief
Applied Mathematics and Computation

Dear Editor,

We submit our manuscript entitled “**Spatiotemporal spectral derivatives with forward stepwise BIC for robust partial differential equation discovery**” by Fuchang Wang and Huirong Cao for consideration as a regular article in *Applied Mathematics and Computation*.

Problem. Data-driven PDE discovery from noisy data faces two practical difficulties. First, high-order spatial derivatives such as u_{xxxx} amplify measurement noise. Second, sparse regression methods like STLSQ require a manually chosen threshold, which varies with noise level and problem. Existing spectral derivative methods are spectrally accurate but assume periodic boundaries; on non-periodic data they suffer Gibbs ringing.

What we do. We propose three connected changes:

1. **Spatiotemporal spectral derivatives.** Instead of transforming space at each time step independently, we apply a single 2D Fourier transform over the joint space-time grid. For a traveling wave $u(x, t) = f(x - ct)$, the spectrum lies along the line $k_t = -ck_x$ (Proposition 2), whereas boundary Gibbs noise spreads isotropically. A 2D filter attenuates boundary artifacts relative to this ridge. This is the key difference from spatial-only spectral methods (Schaeffer and McCalla, 2017).
2. **Spectral weak form.** The regression is set up in weak form: both sides of the PDE are integrated against compactly supported Gaussian test functions. This integration averages out high-frequency noise in the spectrally computed derivatives, particularly for high-order terms. We give a spectral convergence error bound (Proposition 1).
3. **Normalized forward stepwise BIC.** Each candidate term is scaled to unit ℓ_2 norm before regression. Forward stepwise selection with BIC then chooses the support automatically, with no threshold grid search. Under standard restricted-eigenvalue and minimum-coefficient conditions, the global BIC criterion selects the true support with probability tending to one; the forward stepwise search provides an efficient greedy approximation (Proposition 3).

What we find. On the MDBench benchmark (four 1D PDEs at five noise levels, plus three 2D/3D problems):

- On the non-periodic advection equation—where spatial-only spectral derivatives fail completely (0/1 terms recovered)—our method recovers the true term at clean data.
- On Burgers, all true terms are recovered at every noise level including SNR 10.

- On KdV and Kuramoto–Sivashinsky, recovery is achieved at three and two noise levels, respectively.
- On periodic problems, accuracy is comparable to Weak SINDy while runtime is substantially lower.

We compare against three baselines: finite-difference SR3, spatial-only spectral derivatives with exponential filtering, and Weak SINDy. The control-variable comparison with spatial-only spectral methods isolates the effect of the spatiotemporal transform.

Why AMC. The paper combines a numerical algorithm (spatiotemporal spectral differentiation), error analysis (three propositions), and benchmark experiments on a public dataset. It does not require GPU acceleration and runs on a standard CPU. We believe this fits the scope of *Applied Mathematics and Computation*.

This manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the submission. We declare no competing interests. This work was supported by the Langfang Science and Technology Research and Development Program (Grant No. 2024011015). No generative AI was used in the preparation of this manuscript. Code is available at <https://github.com/fzmath/spectral-pde-discovery>.

Thank you for your consideration.

Sincerely,

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